SECTION 7 – RF SHIELDED ROOM

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7–1 RF SHIELDED ROOM SPECIFICATION

Every GE MR system requires that the Magnet Room be RF shielded. Table 7–1 contains the RF Shielded Room specifications.

Note

It is advisable to check the integrity of any RF Shielded (screen) Room in proximity to the MR system. Poor RF Shield integrity can contribute to system image quality artifacts i.e. zippers.

TABLE 7–1
RF SHIELDED ROOM REQUIREMENTS

PARAMETER	REQUIREMENTS
RF ATTENUATION	• 100dB (10MHz – 100MHz) **planewave** with RF vendor supplied blank panel, refer to TESTING requirements in this table.
GROUND ISOLATION	• 1,000 ohms or greater Refer to Section 7–7 ANCHOR SPECIFICATION FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM for additional information on electrical isolation requirement.
MATERIALS	• The choice of material is the responsibility of the customer's architect and RF vendor. Normally, copper–brass or treated aluminum is used because these materials are non–magnetic and will not affect homogeneity. However, RF Shielding has also been fabricated from galvanized steel or by modifying steel magnetic shielding to produce the required RF attenuation. Any steel RF enclosure will affect the magnet's homogeneity and must be reviewed by GE Medical Systems MR Siting and Shielding Group. The door or any other moving or non–rigid parts must not be fabricated from magnetic materials.
CONSTRUCTION	• The design of the shield support system is the responsibility of the customer's architect and RF vendor. If magnetic steel panels are used, these materials must be rigidly supported to prevent any slight movement, from air pressure changes or other reasons, that could degrade magnet homogeneity and system performance. For safety reasons, magnetic steel material must also be well anchored; loose steel components can become dangerous projectiles and accelerated into the magnet. • It is the customer's responsibility to coordinate magnet mounting methods with the RF shielded room vendor to prevent RF leaks and secondary grounding problems.
TESTING	• The customer's architect and RF vendor are responsible for conducting RF attenuation and ground isolation tests to verify that the shield meets GE specifications. The RF shielded room verification test is to be performed in the presence of a GE representative. • When to test: The RF shielded room must be 100% finished, magnet installed, all penetrations complete including floor mounting bolts. The test must be conducted with an RF vendor supplied blank penetration panel and the same mounting hardware which is used with the GE penetration panel. • How to test: The FINAL RF Shielded room acceptance test shall be performed in accordance with Appendix B, RF SHIELDED ENCLOSURE TEST GUIDELINE.
MAINTENANCE	• Follow RF vendor's recommended maintenance. Alert GE Service Representative of any RF shielded room maintenance issues since there may be system performance impacts.
ACOUSTIC	• RF Screen Room including all openings (i.e. windows, doors, vents, etc.) need acoustic properties to meet local regulations and customer requirements. **Note** RF Screen Room doors with <55db acoustic attenuation have caused customer acoustics issues. Refer to Section 4–7 ACOUSTICS for additional information.

7–2 VENTS

7–2–1 Cryogenic

Due to normal boil–off of liquid helium and the possibility of a quench with superconducting magnets, outside cryogenic venting is required. RF shielded room contractor is to provide one straight pipe with maximum 0.125 in. (3.175 mm) wall thickness for the cryogenic vent pipe/waveguide. The vent pipe/waveguide is to be made of non–magnetic material which is grounded to the RF room and electrically isolated from any other grounds. The vent pipe/waveguide must extend inside and outside of the RF shielded room to allow for non–metallic isolation joint connections. The HVAC (heating, ventilation, and air conditioning) contractor is to make cryogenic vent connections to vent pipe/waveguide outside of the RF shield and GE will make the normal connection in the Magnet Room. Refer to Section 4–9–2, Requirements For Inside Magnet Room, for exceptions.

7–2–2 Determining Cryogenic Vent Location

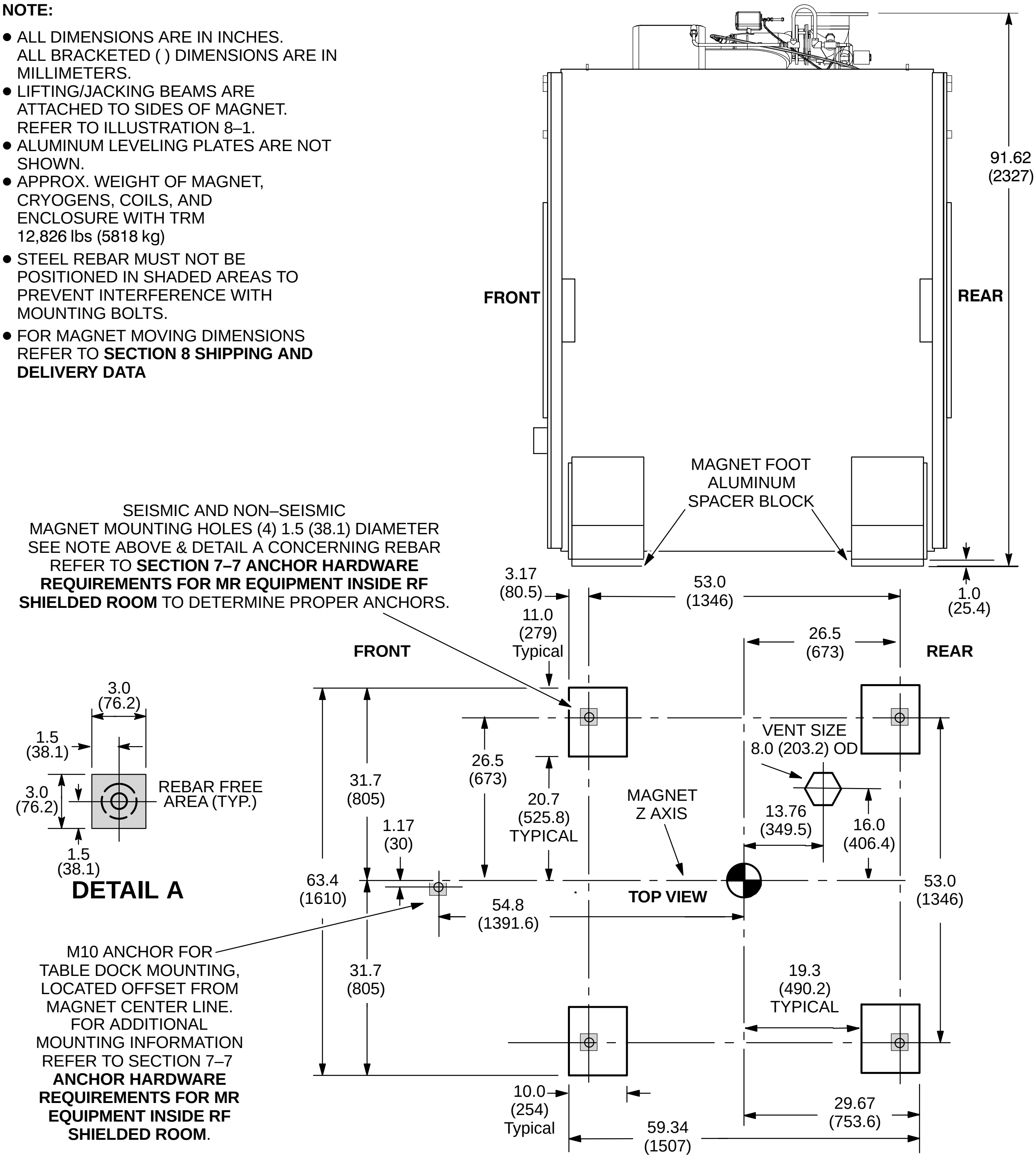
Shown in Illustration 7–1 is the LCC magnet vent location. It is important that the 0.25 in. (6.35 mm) tolerance for vent opening be met. Included with the 1.5T LCC (Active Shield) magnets is a vertical helium vent tube made with with tubing of 6061–T6511 aluminum tubing 8 in. (203.2 mm) OD x 0.125 in. (3.175 mm) wall, per ASTM B241. A typical route for this vent tube is shown in Illustration 7–2. However, if the tolerances shown in Illustration 7–1 can not be met, the customer's contractor is responsible for the design and installation of a cryogenic vent system which meets the requirements in Section 4–9, CRYOGENIC VENTING.

GE provides a Floating Flange Vent Adaptor Kit with all of the 1.5T Active Shield magnet. The kit increases the vent adapter radial adjustability in any direction to one inch (25.4 mm). The adjustability helps absorb some mismatch (1 inch (25.4 mm) in any horizontal direction) of the locations of the Vent Pipe from the magnet and the Ceiling Vent Pipe, installed by the RF Room contractor. See Illustration 7–3. The GE provided Vent Pipe is 24 inches (610 mm) long with a wall thickness of 0.125 inch (3.175 mm) and may be cut short to create 1 $\pm$ 0.25 inch (25 $\pm$ 6 mm) gap between the Vent Pipe and the Ceiling Waveguide for dielectric isolation to ensure the integrity of the RF shield room.

7–2–2 Determining Cryogenic Vent Location (Continued)

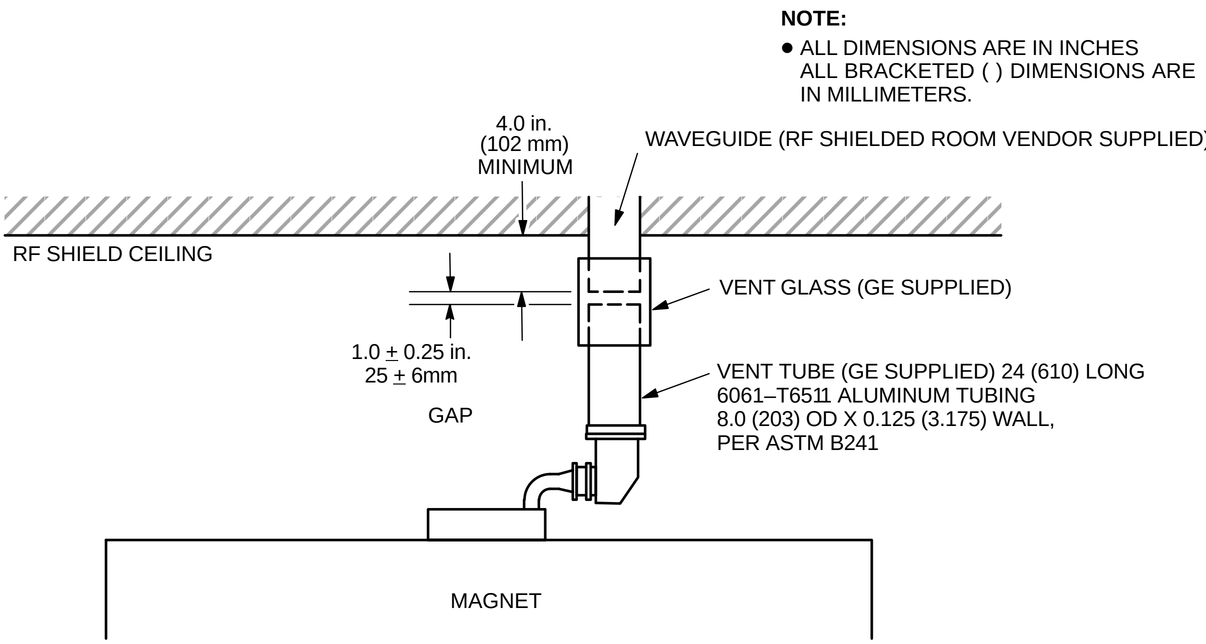
NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- LIFTING/JACKING BEAMS ARE ATTACHED TO SIDES OF MAGNET. REFER TO ILLUSTRATION 8–1.
- ALUMINUM LEVELING PLATES ARE NOT SHOWN.
- APPROX. WEIGHT OF MAGNET, CRYOGENS, COILS, AND ENCLOSURE WITH TRM 12,826 lbs (5818 kg)
- STEEL REBAR MUST NOT BE POSITIONED IN SHADED AREAS TO PREVENT INTERFERENCE WITH MOUNTING BOLTS.
- FOR MAGNET MOVING DIMENSIONS REFER TO **SECTION 8 SHIPPING AND DELIVERY DATA**

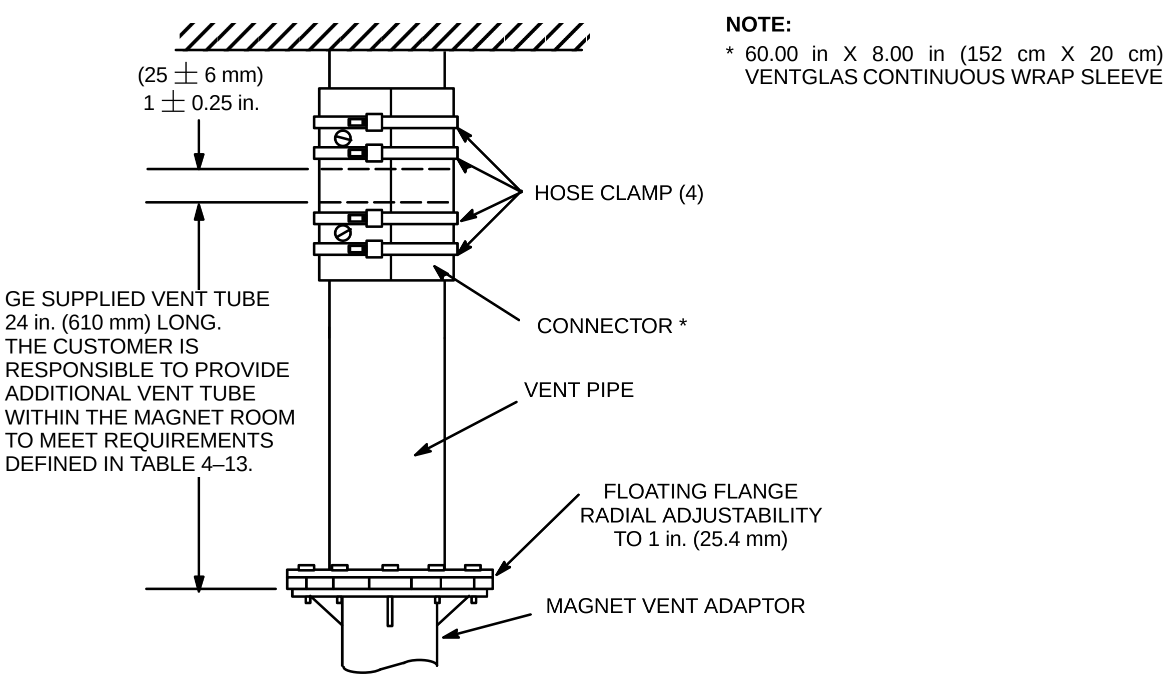


LCC MAGNET CRYOGENIC MOUNTING & VENT LOCATION
ILLUSTRATION 7–1

7–2–2 Determining Cryogenic Vent Location (Continued)



CRYOGENIC VENT ROUTING
ILLUSTRATION 7–2



FLOATING FLANGE
ILLUSTRATION 7–3

7–2–3 Waveguide

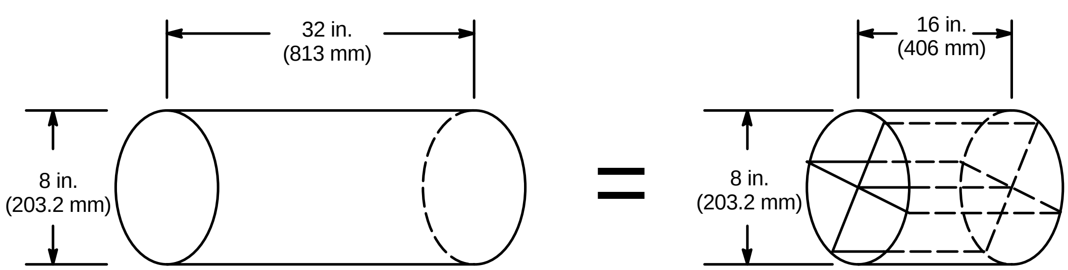
RF shield room contractor/designer is responsible for choosing and installing a RF shield waveguide . The generally accepted length of the waveguide is four times the outside diameter of the tube. Therefore, an 8 in. (200 mm) OD waveguide should be 32 in. (800 mm) long. Refer to Table 7–2 for list of GE requirements for the waveguide.

TABLE 7–2
WAVEGUIDE REQUIREMENTS

PARAMETER	REQUIREMENTS
WAVEGUIDE SIZE	• If the provided GE vent tube is to be connected directly to the waveguide via the GE Ventglas method, the outside diameter of the waveguide must match the outside diameter of the GE vent tube within $\pm$ 0.125 in. (3 mm). Larger mismatches in diameters will result in unacceptable leakage during a quench. • The generally accepted length of the waveguide is four times the inside diameter of the tube. Therefore, an 8 in. (203.2 mm) OD waveguide should be 32 in. (813 mm) long.
WAVEGUIDE MATERIAL	• The Waveguide must be constructed from one of the GE accepted materials (i.e. stainless steel, aluminium or copper). Typically, the waveguide is made from the same material as the RF shielded enclosure to avoid dissimilar metal interfaces (i.e. galvanic reaction).
WAVEGUIDE CONSTRUCTION	• The waveguide does not have to be positioned equally on either side of the RF ceiling (or wall). For example, 6 in. (152.4 mm) of a 32 in. (813 mm) long waveguide may hang down below the RF ceiling with the remaining 26 in. (660 mm) extending above. • If a 90 degree elbow is required to avoid ceiling structures, the elbow can be a part of the waveguide and contribute to its overall length. Waveguides do not have to be completely straight. • If a full length waveguide is not possible due to structural interferences, a shorter waveguide can be fabricated by dividing the inside volume into no more than four chambers. For example, if an 8 in. (203.2 mm) OD waveguide is divided into four equal chambers, as shown in ILLUSTRATION 7–4, the length of the waveguide may be decreased from 32 in. (813 mm) to 16 in. (406 mm). If this technique is used, 1 psig must be added to the pressure drop calculation to account for the pressure drop of the four chambered waveguide. (Section 4–9, CRYOGENIC VENTING) • Flat, honeycomb type waveguide is not acceptable.

NOTE:

- 1 psig MUST BE ADDED TO THE PRESSURE DROP CALCULATION TO ACCOUNT FOR THE PRESSURE DROP OF THE FOUR CHAMBERED WAVEGUIDE.
- IN A CASE OF WAVEGUIDE LENGTH RESTRICTION, A HALF LENGTH WAVEGUIDE WITH FOUR CHAMBERS MAY BE USED.



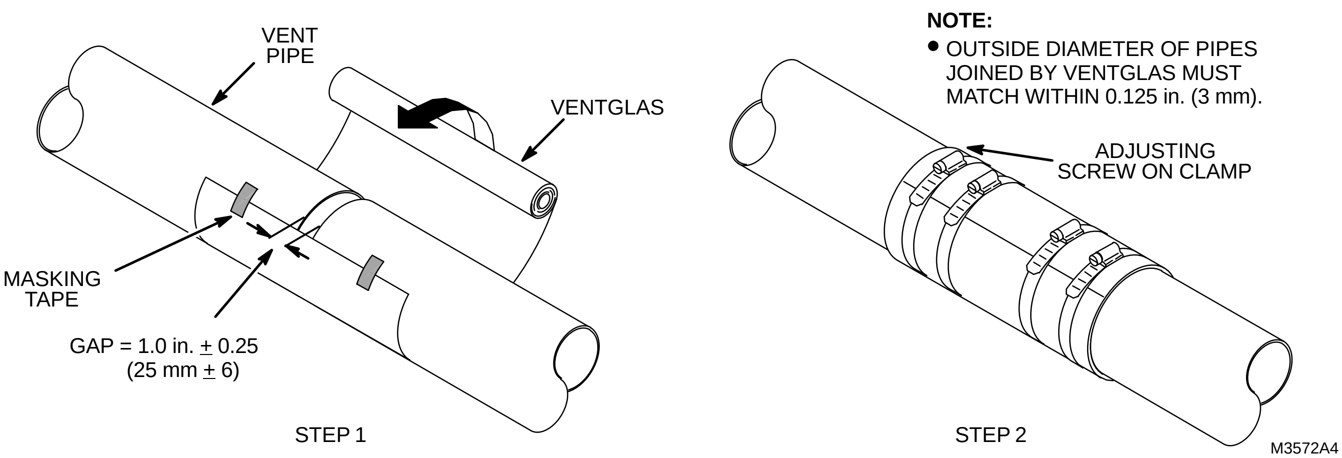
4 CHAMBER WAVEGUIDE
ILLUSTRATION 7–4

7–2–4 Guide For Outside RF Room Isolation Joint

The RF shielded room contractor/designer is responsible for choosing and installing an isolation joint outside of the RF shielded room as shown in Section 4–9–1, Requirements For Outside Magnet Room. This isolation joint is required to maintain the single point ground concept for the RF shielded room. Table 7–3 contains suggestions for the RF room isolation joint.

TABLE 7–3
OUTSIDE RF ROOM ISOLATION JOINT SUGGESTIONS

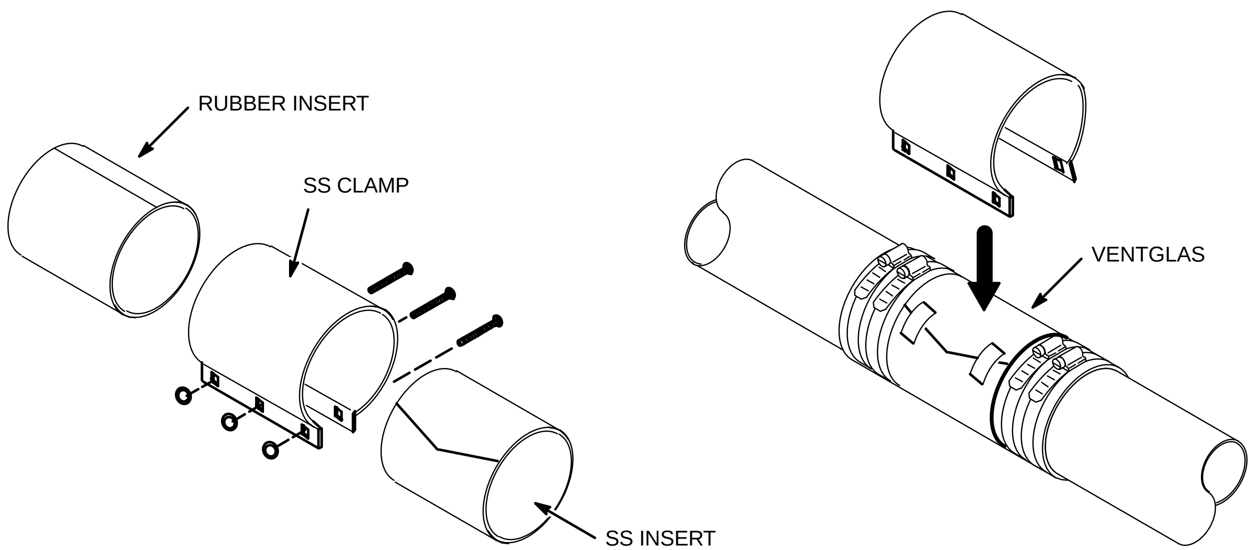
PARAMETER	ISOLATION JOINT SUGGESTIONS
ISOLATION JOINT MATERIAL	• PVC, rubber or soil pipes **must not** be used to construct the isolation joint. • Ventglas and Lorenz clamp are two GE recommended methods of achieving the isolation.
ISOLATION JOINT CONSTRUCTION	• Ventglas connection method is shown in Illustration 7–5 • Lorenz clamp connection is shown in Illustration 7–6 • The mating diameters must match within $\pm$ 0.125 in. (3 mm). • The Ventglas **must not** be used for structural support.
INFORMATION	• Ventglas information may be obtained from: Industrial Machine & Fabricating Inc. e–mail address indmachfab@aol.com 2808 E Sammys Lane Florence, SC 29506–3841 USA (843) 667–4582 Vent Fabrics Inc. web site address www.ventfabrics.com 5520 N Lynch Avenue Chicago, IL 60630–1418 USA (800) 621–1207 or (773) 775–4477 • Lorenz clamp information may be obtained from: Lorenz and Son Mfg. Co. LTD. P.O. Box 1002 Cobourg, Ontario, Canada K9A4W4 (905) 372–2240, fax (905) 372–4456



VENT GLASS CONTINUOUS WRAP
ILLUSTRATION 7–5

7–2–4 Guide For Outside RF Room Isolation Joint (Continued)

NOTE:
- LORENZ CLAMP IS ATTACHED OVER THE VENT GLASS FOR ADDED REINFORCEMENT.
- SS = STAINLESS STEEL



LORENZ CLAMP
ILLUSTRATION 7–6

7–2–5 HVAC

RF shielded room contractor is to install HVAC waveguides (open pipes or honeycomb–type) which penetrate room and to ensure waveguides are non–magnetic and electrically isolated. HVAC contractor is to determine size and number of vents, consistent with local codes.

An exhaust fan placed outside the RF shielding with appropriate wave guide filtering is required for quick removal of helium gas in the event large amounts of helium disperse into the Magnet Room. The exhaust fan can be connected to the output relay of the optional oxygen monitor. The fan will then be activated in the event the room oxygen level is less than 18%. Refer to Section 4–8, ROOM VENTILATION, for other exhaust fan requirements.

7–3 PLUMBING

All metallic pipes entering the RF Room, excluding cryogenic vent and floor drains, must be located within 30 inches (762 mm) of the RF common ground.

Note

When welding in an MR room with system equipment installed, the return path for the welding must be in very close proximity to the welding. The close proximity is needed to make sure the welding currents do not cause damage to the system. Never use the building structure as a return path for welding.

7–3–1 Water

All pipe waveguides must meet the 100 dB requirements. If a floor drain is to be installed in the Magnet Room, it must be electrically isolated and meet the 100 dB requirements. All plumbing must be consistent with local codes.

7–3–2 Medical Gases

The customer should consider if medical gases are to be piped into the Magnet Room along with suction service for patient life support. Remember, all non–electrical entries into the Magnet Room must use appropriate waveguide. Special precaution must be taken to ensure that ferromagnetic medical gas cylinders are not brought into the Magnet Room.

7–3–3 Sprinklers

If using sprinklers in the Magnet Room, dry pipe systems have the advantage of reducing ground problems. However, all decisions regarding fire protection systems are the customer’s responsibility. If wet–type sprinkler system is used, pipe penetration should be limited to one location.

7–4 ELECTRICAL

The entry of any electrical lines into the RF shielded room must be filtered to ensure that the RF shielded room meets the minimum attenuation levels. The RF shielded room vendor must supply filters for all penetrations of the RF shielding excluding the lines entering through the GE supplied RF penetration panel. All filters (for electrical lines) must be located outside the 200 gauss line.

Note

AC outlets must have ground wires firmly attached per electrical codes to avoid intermittent grounding which can cause undesirable emi (electro magnetic interference) issues within the RF shielded room.

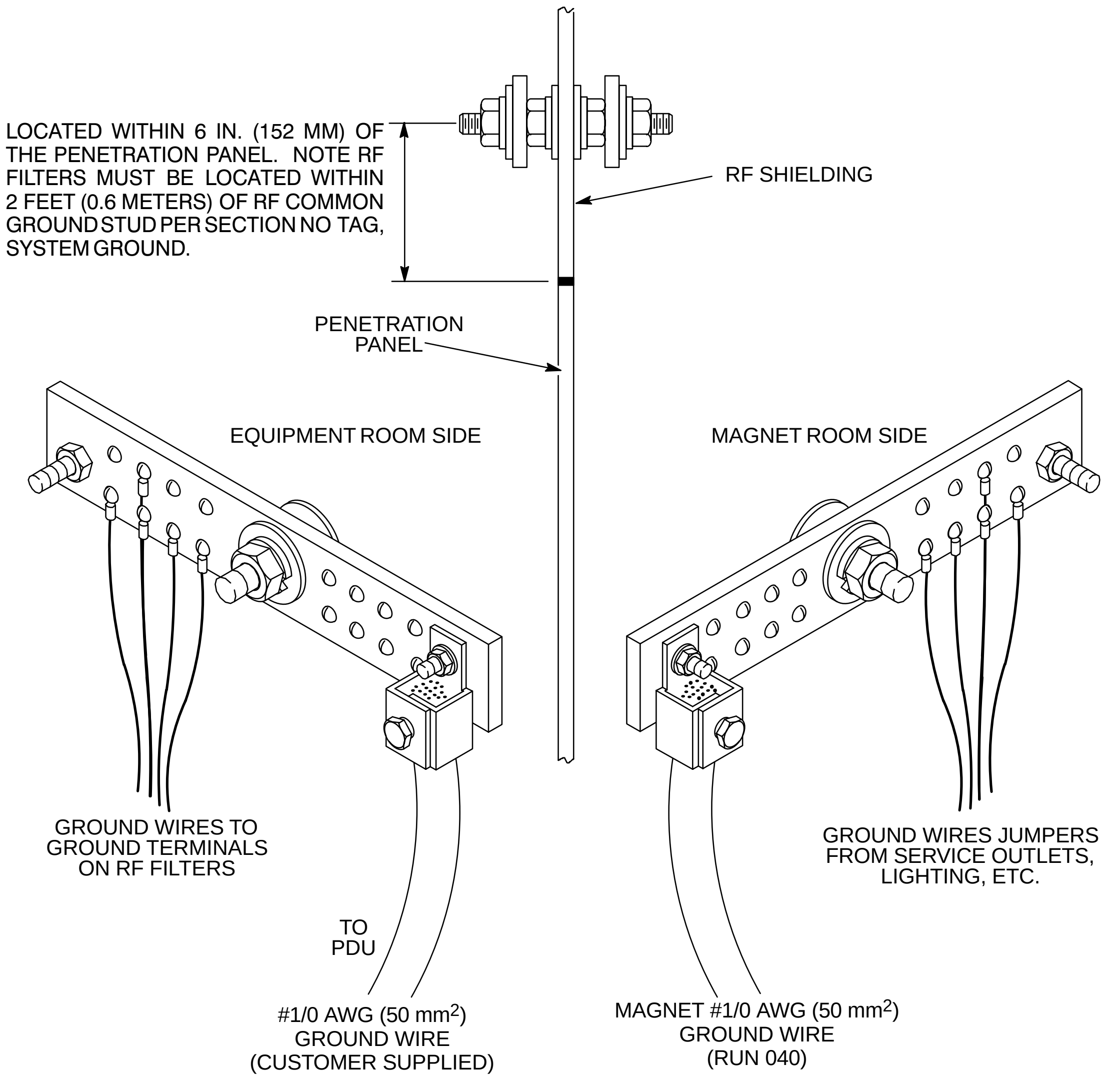
RF shielded room vendor should review with the electrical contractor the number of incoming power lines to the Magnet Room in determining the number of filters needed for electrical requirements.

The power ground lines from customer supplied power filters must be grounded to the RF shield common ground point and be located as close as possible to the RF penetration panel. See Section 5, POWER REQUIREMENTS, for power and grounding requirements of all incoming power lines to the RF shielded room.

All lighting in the RF shielded room must be DC/incandescent. Dimmer switches must not be used; however, a selectable switch may be used to change the light intensity. For additional Magnet Room lighting information refer to Section 4, SITE ENVIRONMENT, Section 5, POWER REQUIREMENTS, and Section 6, INTERCONNECT DATA.

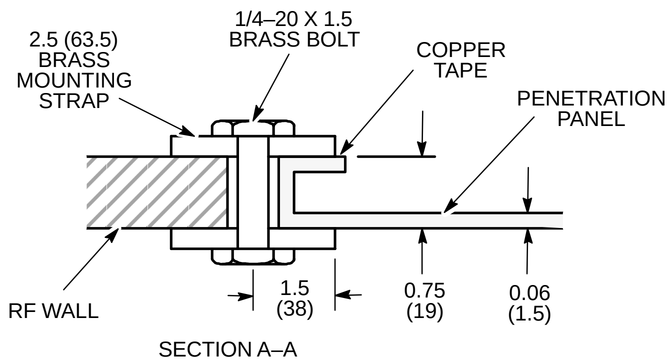
Common ground connection for shielded room must be located within 6 in. (152 mm) of the RF shielded room Penetration Panel with RF filters located within 2 feet (0.6 meters) of the RF common ground. RF shielded room vendor to provide this common ground connection on both sides of shielded room by means of a stud extending through the shielded room (see Illustration 7–7). RF Common ground stud must be accessible for servicing purposes on both sides of shield room. For aesthetic purposes, it is recommended that the stud be positioned above the Penetration Panel so it is concealed behind the penetration panel cover (see Illustration 7–10).

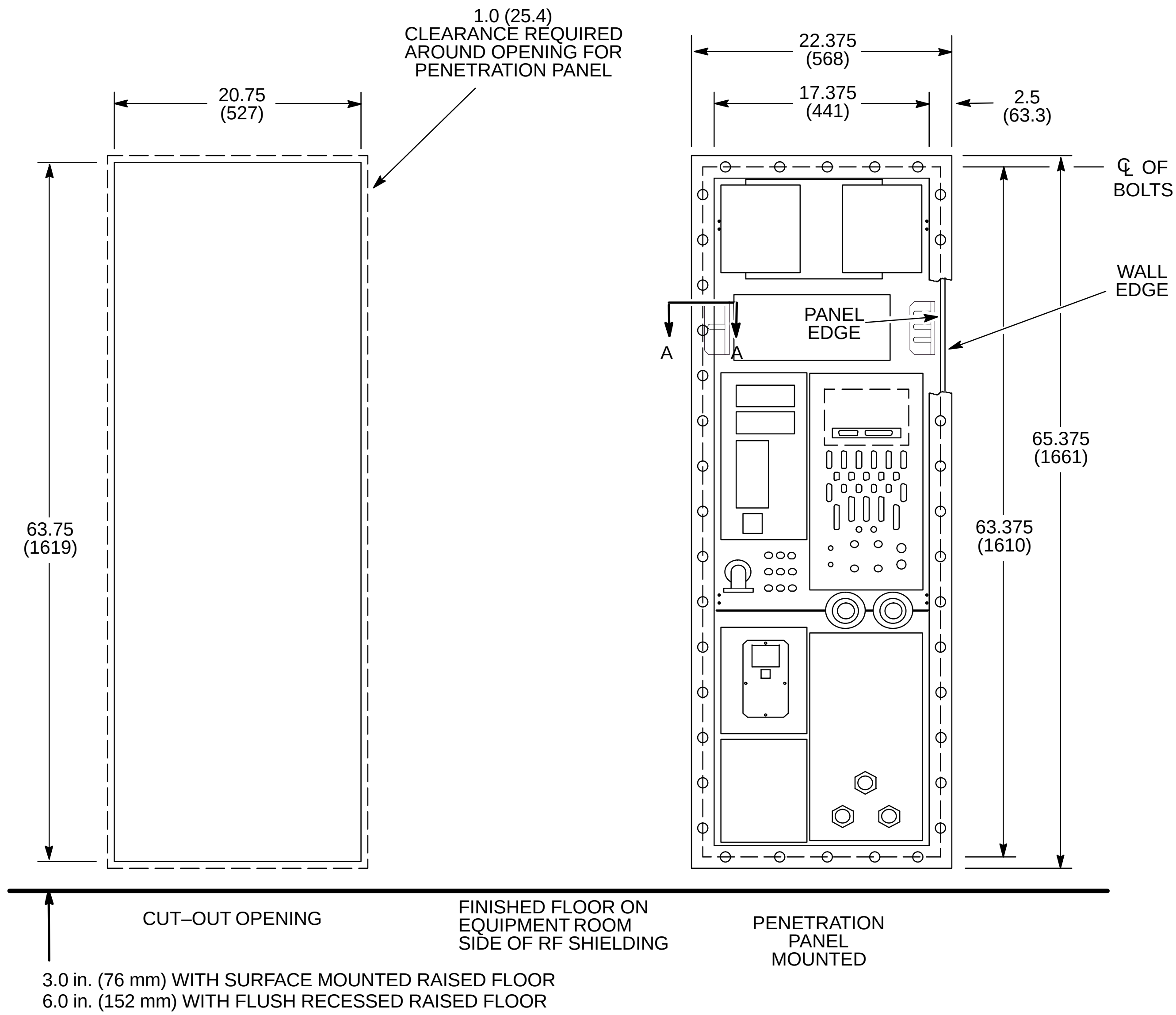
NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS



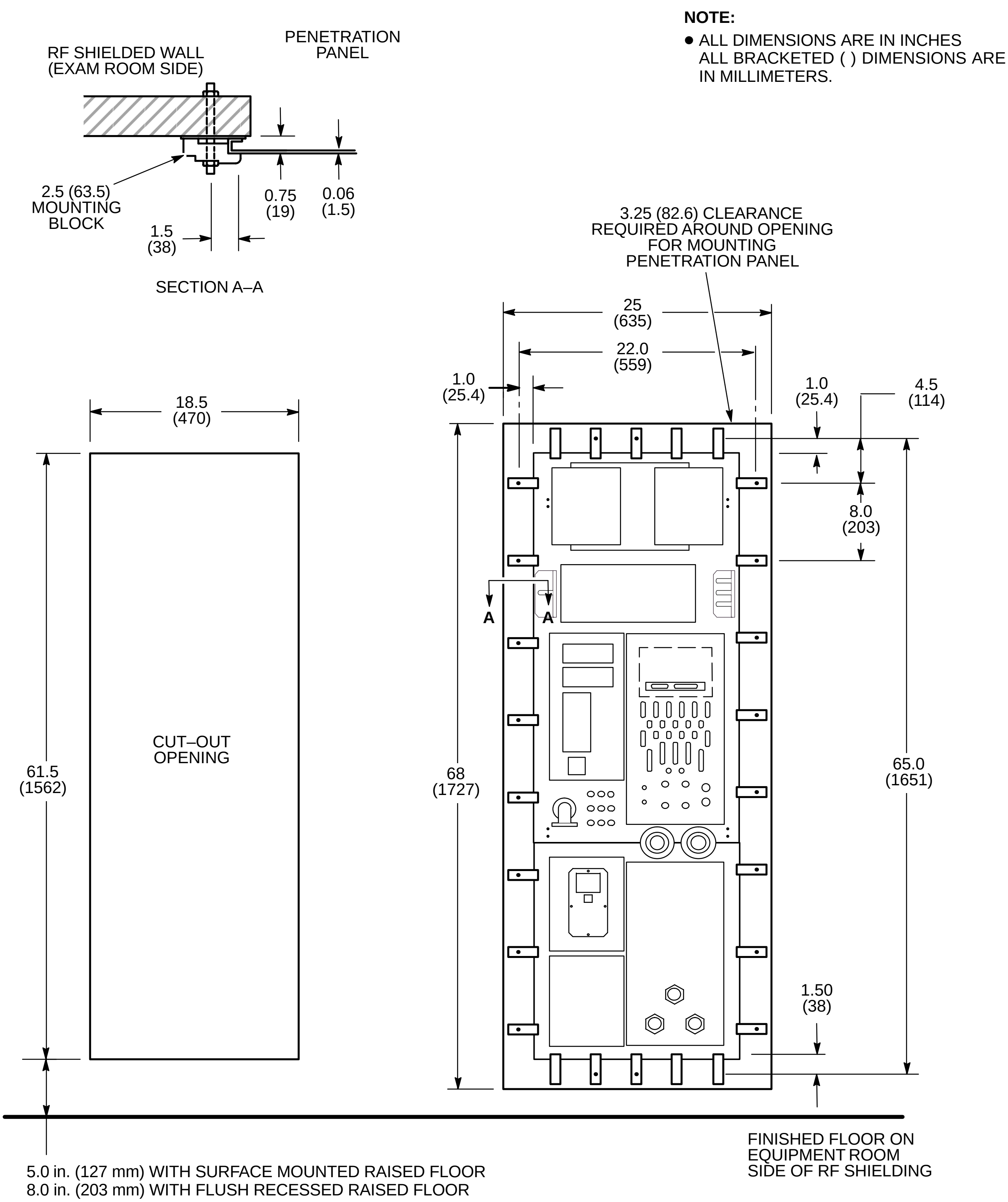
RF COMMON GROUND PENETRATION STUD
ILLUSTRATION 7–7

NOTE:
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- ALL DIMENSIONS ARE ± 0.0625 (1.6)





PENETRATION PANEL CUT OUT FOR 0.75 INCH (19 MM) THICK RF WALL
ILLUSTRATION 7–8



PENETRATION PANEL CUT OUT FOR RF WALL THICKNESS VARYING FROM SMALL TO LARGE
ILLUSTRATION 7–9

7–5 RF PENETRATION PANEL

Illustrations 7–8 and 7–9 show two different methods for mounting the GE MR penetration panel. Either method may be used depending on RF shielded room wall thickness. Ensure if the mounting method in Illustration 7–8 is used, the RF wall thickness is 0.75 in. (19 mm) $\pm$ 0.0625 in. (1.6 mm). Check with RF shielded room vendor to determine appropriate mounting method. The penetration panel must be covered on both sides for safety. If GE supplied adjustable covers are not used, customer must furnish covers or enclosures with key or tool required for opening to limit access to the panel.

The penetration panel is to be mounted above the finished floor on the penetration cabinet side of the RF shielded room. GE supplies only the penetration panel as shown in Illustrations 7–8 and 7–9. The mounting and clearance dimensions for the penetration panel covering are shown in Illustration 7–10.

RF shielded room vendor must provide the opening in the RF shielding and appropriate mounting hardware for the GE penetration panel. The RF shielded room acceptance test must be performed after the opening is cut in the RF shielding for the GE penetration panel. This acceptance test must be conducted with vendor supplied blank panel and the same mounting hardware to be used with the GE penetration panel. It is the facility's responsibility to ensure that the RF shielded room vendor testing meets the attenuation specifications listed in Section 7–1, RF SHIELDED ROOM INTRODUCTION.

7–6 PHYSICAL CONSIDERATIONS

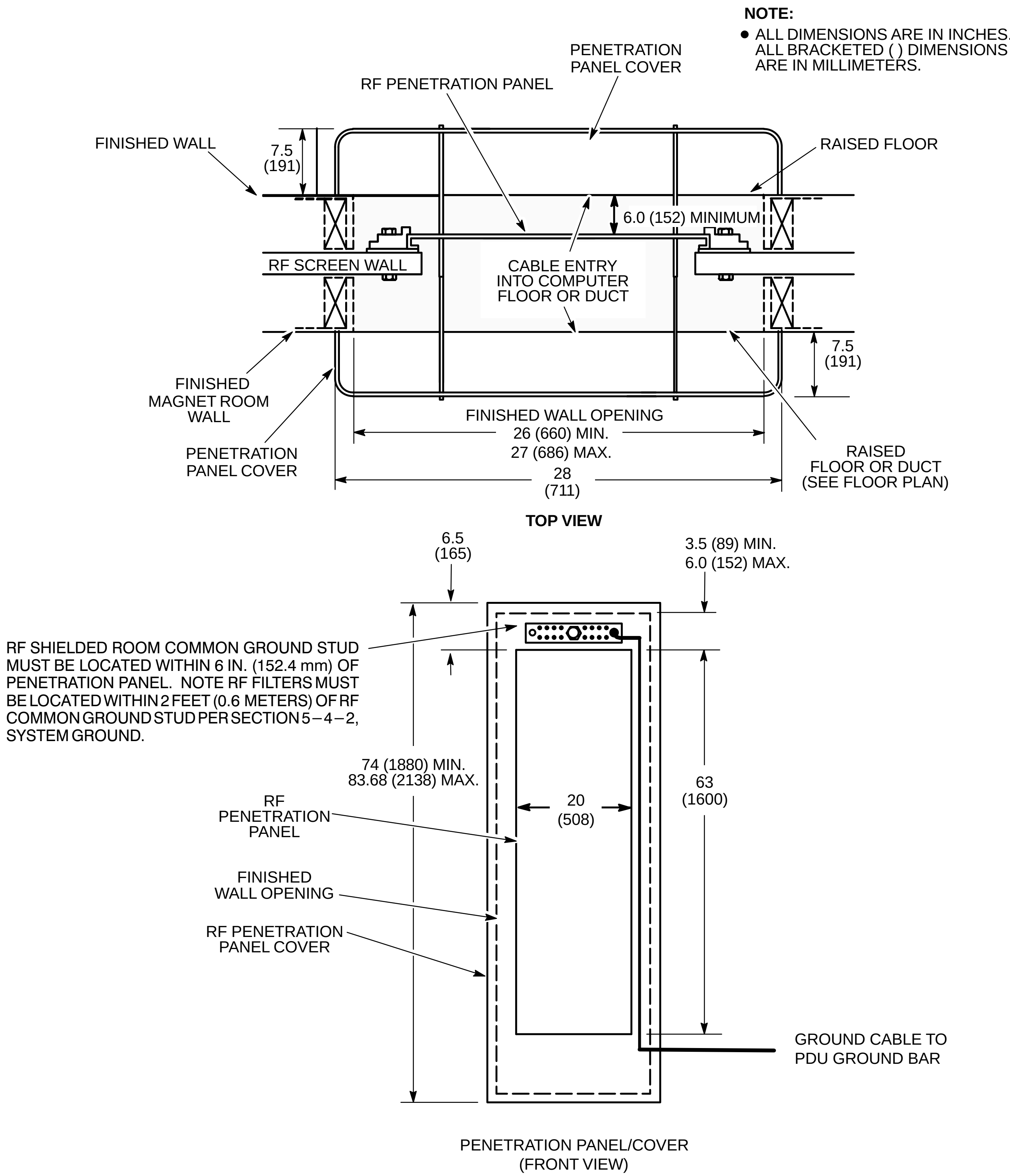
The RF shielded room can be either a free standing shielded structure or a shielded room within an existing room. Either style must be electrically isolated from earth ground by 1000 ohms or greater. Magnetic shields that are used as RF shield must also be isolated from ground by the same specification.

7–6–1 Doors and Other Openings

Shielded room doors are a major source of RFI leaks and should be kept to a minimum.

The main door requires a minimum finished opening of 43 in. (1092 mm) to allow for helium dewars and patient tables to pass through the opening. However, a 48 in. (1219 mm) wide door is recommended for easy maneuvering of the patient table. Maximum door sill height is 1 in. (25 mm) with a 10 degree maximum threshold inclination.

For moving the standard magnet into the room, a 9 ft wide by 10 ft high removable panel wall or 9 ft by 10 ft roof hatch is required. This normally consists of several panels that are bolted together to ensure a good RF shield when opening is not in use. Since future access may be required for magnet entrance/exit through the removable panels, avoid permanently sealing the removable panels. Note the room removable wall panel or roof hatch opening may need to be larger to achieve the required opening defined above in this section. Consideration for clear opening dimensions is especially applicable to sites requiring magnetic shielding.



PENETRATION PANEL/COVERING MOUNTING REQUIREMENTS
ILLUSTRATION 7–10

7–6–2 Ceiling Height

Table 7–4 lists the Magnet Room absolute minimum ceiling height required for servicing the listed magnets. This height is required for the area directly above the magnet. GE Medical Systems, Modality Installation Planning Service group must be notified of any ceiling dimensions less than ceiling heights stated in Table 7–4 and shown in Illustration 7–11. Modality Installation Planning Service group can be reached at (262) 548–4500.

TABLE 7–4
MAGNET SERVICING CEILING HEIGHT REQUIREMENTS

MAGNET TYPE	ABSOLUTE MINIMUM CEILING HEIGHT IN. (MM) See Note 1	COMMENTS
1.5T LCC (Active Shield)	105 (2667)	Magnet servicing is performed from a platform ladder which is positioned at the Coldhead side of the Magnet. Ceiling height allows for clearance for fill line stinger insertion.
1.5T LCC (Active Shield) with Low Ceiling Height Siting Option (M1060SR) installed	98.5 (2500)	Magnet servicing is performed from a platform ladder which is positioned at the Coldhead side of the Magnet. Ceiling height allows for clearance for insertion of fill line with 7 inch (178 mm) stinger insertion.
Note * Magnet servicing is performed on a platform mounted on top of the Magnet. Ceiling height includes 39 inches (991 mm) of clearance for Service Personnel. 1 Absolute minimum ceiling height values are from magnet room finished floor to fixed ceiling.		

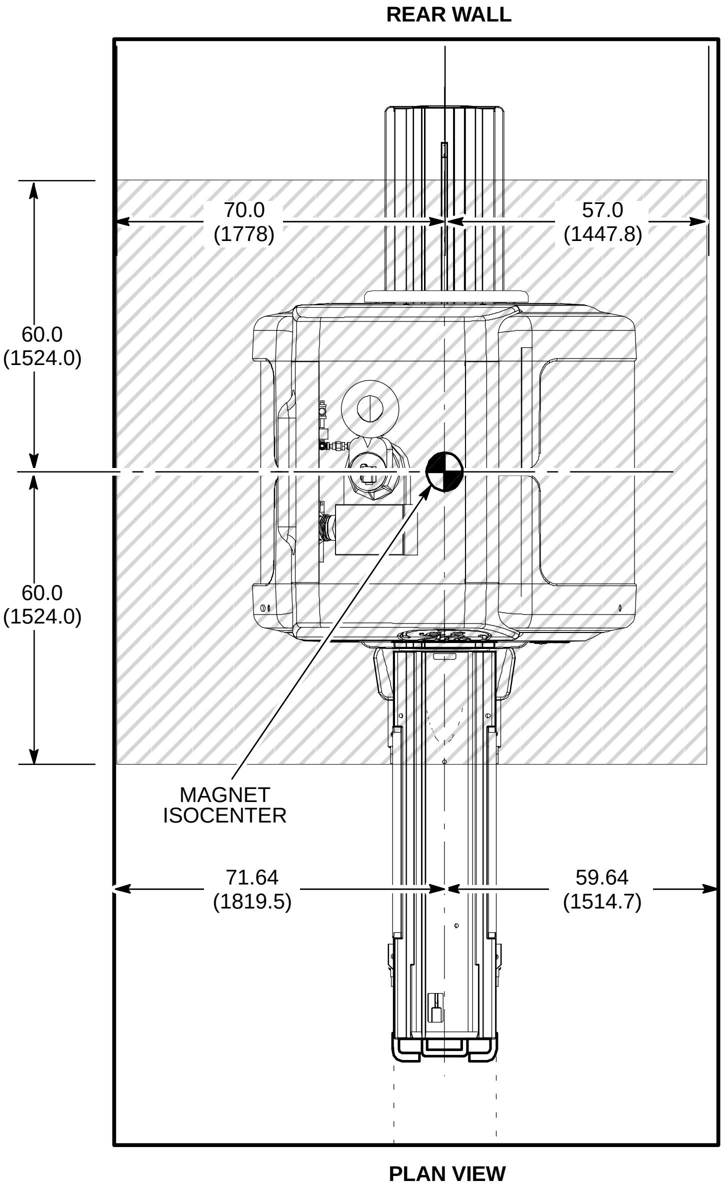
Use of a standard valved helium transfill line and a 250 liter dewar (not more than 70 in. (1778 mm) high) requires a ceiling height of 135.5 in. (3442 mm) for inserting transfill line into the dewar. Note that this need only be a 24 in. (610 mm) square ceiling recess located either in the Magnet Room or in an accessible area near the Magnet Room where the transfill line can be inserted into the dewar. A 500 liter dewar (not more than 73 in. (1854 mm) high) requires a ceiling height of 138 in. (3505 mm) for the same process.

If the helium transfill requirements cannot be satisfied in or near the Magnet Room, consider a location outside the building or on a loading dock. The standard valved transfill line, after insertion into either a 500 or 250 liter dewar, will fit through 79 in. (2007 mm) high doorways and hallways. Provide free access from the dewar location to the magnet. If elevators are to be used along cryogen delivery route, verify that elevator dimensions and weight capacity is sufficient to handle the cryogen dewars. Also, elevator must be dedicated with restricted access during cryogen transport (will not allow stops between initial start and final floor destination).

7–6–2 Ceiling Height (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
  ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- SHADED AREA ROOM CEILING MUST NOT BE BELOW TABLE 7–4 REQUIREMENTS.



MINIMUM CEILING HEIGHT 1.5T LCC (ACTIVE SHIELD) MAGNET
ILLUSTRATION 7–11

7–6–3 Walls

It is recommended that walls be covered to protect RF material and to add to the aesthetics of the room for patient comfort. Fire retarding material must be used per building codes. Consult RF shield room vendor for RF shielding service requirements prior to covering RF walls. Removable wall covering may be needed if periodic RF shield servicing is required to maintain RF integrity.

The recommended patient viewing window dimensions are 48 in. wide by 42 in. high (1219 mm x 1067 mm). The location of the window is dependent on the position of Operator Workspace position.

Note
The operator at the Operator Workspace must be able to view the patient during a scan.

7–6–4 Floors

The Magnet Room floor levelness requirement is important for accurate patient table docking. Floor levelness in the Magnet Room must not be greater than 0.3125 in. (8 mm) between depression and high spots over any 120 in. (3048 mm) distance within the area of the magnet enclosure and the area in front of the enclosure, see shaded area shown in Illustration 7–12.

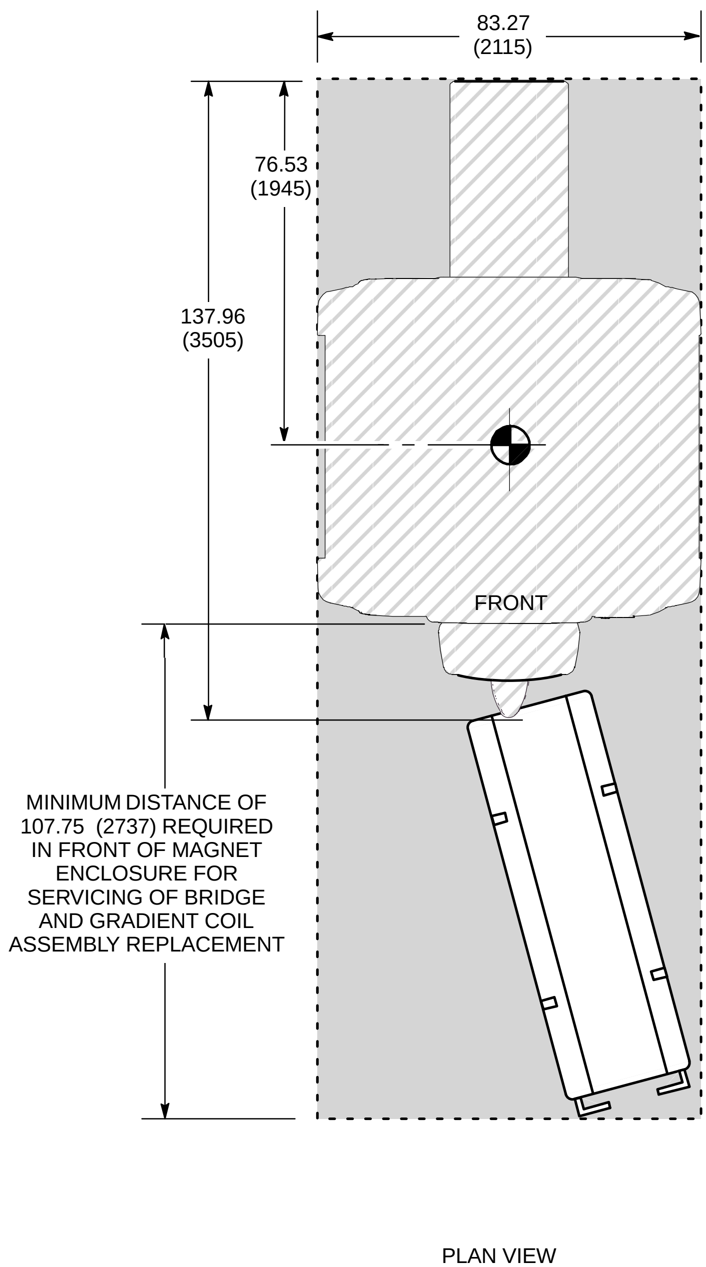
The finished floor is to be designed to insure the finished floor to the center of the magnet dimension of 42.125 in. $\pm$0.25 in. (1070 mm $\pm$6.35 mm) is maintained. This will allow the Patient Transport to properly dock to the enclosure. Also the material used for the finished floor directly under the magnet must not allow the magnet to migrate into the material therefore requiring constant leveling (e.g. vinyl composition tile over a 0.5 inch self leveling top coat).

The finished floor in the Magnet Room should be waterproof and be a conductive type flooring to reduce the possibility of a static discharge. The floor covering should be hard–surfaced with the seams sealed to protect the RF flooring against possible water damage. This floor should also be easily replaceable to repair tiles that are damaged from occasional cryogen spillage. Removable access flooring in the Magnet Room must be aluminium or non magnetic for safety purposes.

Note
The Magnet Room floor must be designed to support TRM Gradient Coil replacement. Refer to **Section 2–4 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY** for weight and dimensions of the TRM Gradient Coil on cradle/cart utilized for replacement.

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- ROOM FLOOR MUST BE CLEAR AND LEVEL FLOOR TO THE ROOM DOOR TO ALLOW FOR PATIENT TABLE MOVEMENT.



MAGNET FLOOR LEVELNESS AREA
ILLUSTRATION 7–12

7–7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM

Overview & Background Information

All GE MR systems require an RF enclosure which is commonly called an RF Shielded Room or an RF Screen Room. RF signals from sources outside of the RF Shielded Room are attenuated so they do not interfere with the MR system. Likewise, the RF signals produced by the MR scanner are kept from interfering with other RF devices. This RF quiet environment is necessary for the MR system to produce quality images.

RF Shield Rooms come in a variety of shapes and sizes but the feature they all have in common is a total RF shield produced by one continuous ground plane. This is achieved by making the walls, floor, doors and ceiling out of an electrically conductive material such as copper, aluminum, brass, or steel. All the room components (walls, floor, ceiling, etc.) must be electrically bonded together to form one solid, common ground plane. Once this is established the ground plane is then tied to earth ground to create the RF shield. This ground point is known as the PRIMARY ground. The primary grounding technique works well for any battery powered devices that may be operated within the RF Shield Room. Devices requiring facility power or the introduction of facility power into the Screen Room requires a change in the primary grounding technique for the RF Shield Room. The RF Shield Room must now be grounded back to the facility power ground.

The addition of water systems or other grounds required by the national electrical code will cause the RF Shield Room ground impedance to be ZERO ohms. These additional grounds that connect the outside of the RF Shield Room to earth grounds are called SECONDARY grounds. It is the secondary grounding that needs to be controlled. If the secondary ground introduces any current to the RF Shield, indicating the RF Shield is a better ground path than the secondary ground, then the current can set up electrical fields on the surface of the RF Shield which may cause image artifact.

The equipment anchors are generally installed into the grade below the RF Shield Room because the floor thickness within the room does not provide the proper embedment depth. With this in mind the hardware used to anchor the equipment will be either a bolt or a stud. This hardware must penetrate the RF Shield and be in full contact with the shield thereby preventing RF leaks at the point of penetration as specified in **Section 7–7–2 RF Shield Integrity**. In this case, the anchoring hardware may be in direct electrical contact to the anchored equipment. The anchoring device that is set below the RF Shield Room floor must be isolated from ground. If it should come in contact with rebar or wire mesh and this in turn is contacting building steel then a secondary ground has been established and needs to be corrected.

RF Shield Rooms that have the RF shield far enough below the finished floor in the room to allow the anchor to be installed without penetrating the shield have no issues with secondary grounding.

Following are the detailed roles and responsibilities of the Customer, RF Shield Room vendor and GE Medical Systems.

The Customer is responsible for the following tasks:

- Coordinate equipment anchor methods and anchor location with the contracted RF shield room vendor, structural engineer, and architect to prevent RF leaks and secondary grounding problems.
- Coordinate with the contracted RF shield room vendor, structural engineer, and architect the design of the equipment anchor hardware per GE specifications, refer to Section 7–7–6, Example – Select Magnet Anchor Size.
- Obtain any and all approvals necessary for the construction of equipment support and seismic anchoring. Provide copy of building inspector's (inspection) report and approval on anchor method to GE Medical Systems local office.

7–7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM (Continued)

The RF Shield Room vendor is responsible for the following tasks:

- Assist Customer structural engineer to determine stud length due to floor and room shield thickness variability.

WARNING!

THE RF SHIELD ROOM VENDOR MUST SUPPLY TORQUE SPECIFICATIONS FOR ALL ANCHORS, STUDS, AND FASTENING HARDWARE TO MEET THE CLAMPING FORCE SPECIFIED EACH PIECE OF EQUIPMENT LISTED IN TABLE 7–5.

- Procurement of commercially available anchors, studs, and fastening hardware required for equipment listed Table 7–5.
- Layout and installation of the equipment anchors (create own template from GE supplied information) Coordination with Building Contractor/Architect may be necessary to prevent interference with rebar or structural steel that would cause a secondary ground path through the anchor.
- GE Service to assist RF Shield Room vendor by locating magnet isocenter during layout of equipment anchors. RF Shield Room vendor to be present when GE Service identifies magnet isocenter location to maximize the accuracy of the location.
- Anchors for Magnet shall be installed prior to magnet delivery to allow time to address any issues that may arise.
- Acnhor for Dock Assembly shall be installed during installation process of the Dock Assembly to the Magnet.
- RF Shield Room Vendor to perform pull test on all anchors and indicate the torque required to achieve the specified clamping force (tension).
- Perform ground impedance test on installed anchors prior to connection to RF shield
- Perform RF integrity test on room after anchors and studs are installed and connected to RF shield
- Provide copy of ground impedance and RF room integrity tests to GE Medical Systems local office
- Required to use a electrically conductive fiberous or equivalent washer to electrically connect anchors and studs to the RF Shield. Flat washers will not maintain RF Shield integrity. Flat washer may introduce image artifacts.

7–7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM (Continued)

GE Medical Systems is responsible for the following tasks:

- GE Installation Specialist to provide equipment dimensions drawings showing equipment mounting locations to the RF Shield Room vendor and review the anchor method prior to anchor installation.
- GE Service to assist RF Shield Room vendor by locating magnet isocenter during layout of equipment anchors. RF Shield Room vendor to be present when GE Service identifies magnet isocenter location to maximize the accuracy of the location.
- GE Service to inspect and verify the anchor location is correct and obtain the anchoring hardware from the RF Shield Room vendor prior to equipment delivery. Carefully inspect the electrical connectivity of the anchor/stud to the RF Shield to make sure the fiberous washer or equivalent device is in place.
- GE Service to work with riggers to secure Magnet to anchors at time of delivery and installation.
- GE Service to attach/align Dock Assembly to Magnet and mark anchor location for RF Shield Room Vendor.

Refer to Table 7–5 for equipment type and seismic anchor characteristics.

7–7–1 Clamping Force (Tension) And Pull Test

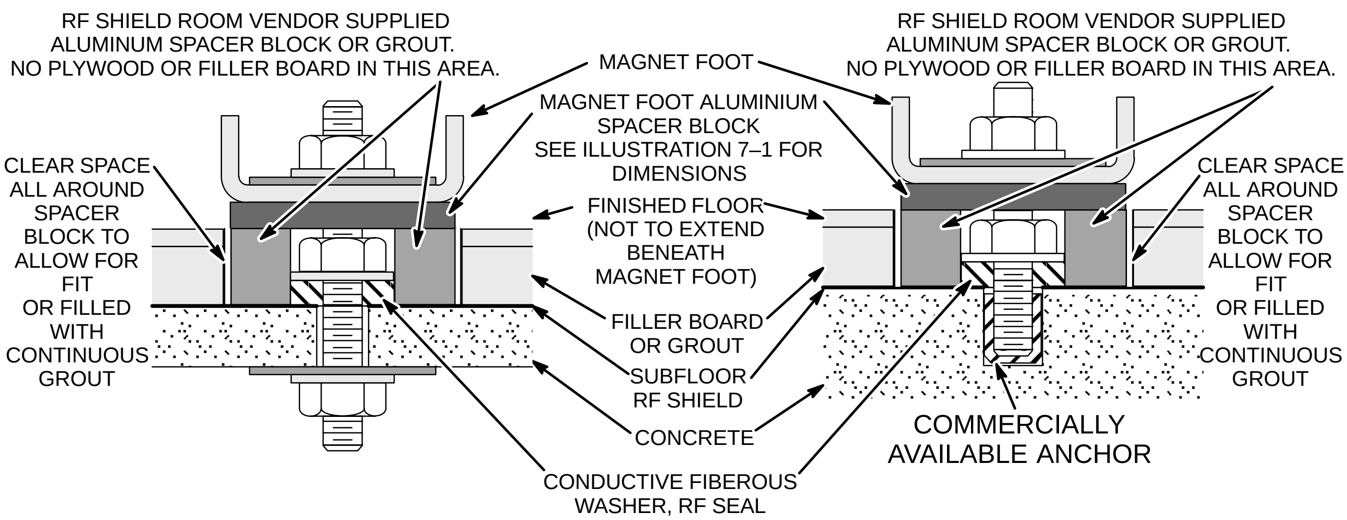
1. The type of anchor to be used for mounting the equipment will be dependent on the building construction (refer to Illustration 7–13) and the clamping force (tension) requirement for the equipment to be anchored (refer to Table 7–5 for equipment clamping force (tension) requirements).

2. To understand how to properly select the anchor diameter and embedment refer to Section 7–7–6, Example – Select Anchor Size.

3. Each anchor must meet the clamping force (tension) requirement.

4. The equipment must be clamped directly to the floor and the entire equipment base must maintain full contact to the floor.

5. A pull test equal to the clamping force (tension) must be performed by the RF Shield Room Vendor prior to the equipment delivery. The test results indicating the torque required to achieve the specified clamping force (tension) must be recorded by RF Shield Room Vendor and the information forwarded to GE Medical Systems for the customers record file.



Note: For sites with RF Shield on top of subfloor, the RF Shield needs to be recessed to the concrete level to provide a proper RF Seal.

RF SHIELD ROOM ANCHOR DETAILS
ILLUSTRATION 7–13

7–7–2 RF Shield Integrity

The anchor hardware must maintain RF shield integrity. This is accomplished by electrically sealing the stud at the penetration point on the RF shield. The method by which the electrical contact is made must take into account any stretch in the stud resulting from the applied clamping force (tension). A fiberous washer or equivalent will provide a proper RF seal where a solid flat washer could produce an RF leak and introduce artifacts into the MR images. The RF room test should result in a specific attenuation at the operating frequency of the system under the following conditions:

1. Blank Penetration Panel installed
2. Anchor hardware installed
3. Electrical connection made between the anchor stud and the RF shield.

Refer to APPENDIX B – RF SHIELDED ENCLOSURE TEST GUIDELINE.

TABLE 7–5
EQUIPMENT CLAMPING FORCE (TENSION) REQUIREMENTS

EQUIPMENT TYPE	**CLAMPING FORCE (TENSION) TO THE FLOOR APPLIED TO EACH ANCHOR lbs (N) See Note 1**
1.5T LCC (Active Shield) Magnet (See Note 2)	2,500 ± 200 (11,100 ± 900)
Dock Assembly for LCC (Active Shield) Magnet	600 ± 100 (2700 ± 450)
Blower Box (See Note 3)	100 ± 10 (450 ± 45)

Note 1 The RF Screen Room Vendor must perform a pull test on each anchor, equal to the clamping force (tension) required for the equipment, prior to equipment delivery. A copy of the pull test results, anchor ground impedance measurements and building inspection certification must be given to GE Medical Systems for customer files.
A floor made of compressible material may require the anchor hardware to be tightened several times to achieve the clamping force (tension). The clamping force (tension) must be maintained for the equipment over the life of the product.

2 All four feet of the magnet must be anchored to the floor. The bolt hole openings in the magnet base are to be used to anchor the magnet.

3 The Blower Box contains magnetic material and must be securely mounted to the floor of the Magnet Room or a access support shelf on the Magnet Room wall or ceiling with support provided under the box. RF Shield integrity must be maintained for mounting the Blower Box within the Magnet Room. The Blower Box may be mounted to a raised floor section within the magnet room. If the Blower Box is secured to the finished floor and the anchor hardware penetrates the RF shield then the RF Shield Vendor must install the anchors prior to system equipment delivery. The need for Blower Box anchors must be planned in advance to allow the RF Shield Room Vendor to include the cost in their bid.

7–7–3 Electrical Isolation

The anchor hardware must not provide a secondary ground path for the RF Shield Room. A secondary ground may occur by having the anchor come in contact with steel rebar, wire mesh or structural steel in the floor. Ideally the ground impedance between the anchor and earth ground should be greater than 1000 ohms. This may not be possible due to moisture conditions in the concrete or soil beneath the concrete. Therefore the following requirements for ground isolation shall be observed:

1. A ground isolation test performed on each electrically conductive anchor and stud.
2. If the result is greater than1000 ohms on each stud then record the resulting measurement and give documentation to GE Medical Systems.
3. If the result is less than1000 ohms but greater than100 ohms, contact the power and grounding support personnel at GE Medical Systems and review process and site conditions.
4. If the result is less than100 ohms then it is very likely the anchor has made contact to steel rebar or wire mesh. In this case the steel in the floor will need to be removed or the anchors will need to be relocated. In either case GE Medical Systems must be notified and a retest performed after the corrective action is taken.
5. The test results must be recorded by RF Shield Room Vendor and the information forwarded to GE Medical Systems for the customers record file.

Electrical Isolation Measurement Method

Prior to attaching the primary ground and installing any power outlets, room lights, water supplies, etc. into the RF Shield Room, a ground isolation test is performed between the room's ground plane and earth ground.

- This measurement should be greater than 1000 ohms DC. However, the results of this measurement may appear lower for RF Shield Rooms located below or at grade level. This is caused by a capacitive voltage, setup by the measurement meter's DC voltage, between the RF Shield Room and earth ground.
- For RF Screen Rooms with a resistance reading less than 1000 ohms the following method should be used to determine if the low resistance reading is caused by this capacitive effect.
  - Use a Meggar Insulation Tester capable of reading < 1000 ohms to most accurately make the measurement. An analog meter with a D'Arsonval meter movement may be used if a Meggar is not available.
  - After making the measurement reverse the leads and watch for the measurement to start high and decrease to a lower resistance value. This change in measurement verifies the capacitive effect.
  - In this case the peak measurement is the approximate resistance between the RF Shield Room and earth ground.

7–7–4 Physical Characteristics

Stud Size

The following factors will contribute to the stud size:

- Magnet anchor/stud diameter: minimum 0.625 in. (M16) and maximum 1.25 in. (M32).
- clamping force (tension) requirement for the equipment, refer to Table 7–5
- Seismic codes (affect the length and diameter of the stud and anchor size)
- Embedment depth (affects the length of the stud)
- Variation in floor thickness (RF Shield Room floor and finished floor)
- Equipment base thickness and any spacers required under the base
- Equipment base clearance for protrusion of stud inside the base
- Size of the hole in the equipment base (affects the diameter of the stud)

Refer to Table 7–6 for Equipment Characteristics.

TABLE 7–6
EQUIPMENT CHARACTERISTICS

EQUIPMENT TYPE	**EQUIPMENT MOUNTING BASE THICKNESS in. (mm)**	**CLEARANCE HOLE IN EQUIPMENT BASE in. (mm)**	**MAXIMUM CLEARANCE ABOVE EQUIPMENT BASE in. (mm) See Note 1**	**SEISMIC REQUIREMENT PRE–APPROVED BY OSHPD FOR BOLT OR STUD DIAMETER in. (mm) See Note 2**	**EQUIPMENT MOUNTING ILLUSTRATION**
1.5T LCC (Active Shield) Magnet See Note 3	1.75 (44) See Note 4	1.5 (38)	2.5 (64)	1.0 (M24)	7–1
Dock Assembly for LCC Magnet	0.75 (20)	0.43 (11)	2.0 (50)	Pre–approval not required. Select anchor/stud to clamping force (tension).	7–1
Blower Box See Note 5	0.25 (6)	0.25 (6)	0.5 (13)	Pre–approval not required. Select anchor/stud to clamping force (tension).	2–27

Note 1 Maximum Clearance Above Equipment Base is the dimension for protrusion of the stud inside the equipment base including clearance for tools to tighten hardware to meet specifications in Section 7–7–1 Clamping Force (Tension) And Pull Test.

2 Seismic codes do not allow for any gap between the stud and clearance hole in the base of the equipment. In California, USA all anchor hardware designs must be submitted to the Office of Statewide Health Planning and Development (OSHPD) for pre–approval. For all other states, providences or countries, plans must be submitted to the local planning authority (where seismic codes apply).

3 All four feet of the magnet must be bolted to the floor. The bolt hole openings in the magnet base are to be used to anchor the magnet.

4 LCC Magnet mounting base thickness includes the 1 inch (25 mm) foot block required to maintain the Magnet Center Line Height dimension as defined in Illustration 2–9.

5 The Blower Box for the Signa Horizon system can be mounted to either the wall or the floor. The floor mounting could be made onto a raised floor section within the magnet room. If the Blower Box is secured to the finished floor and the anchor hardware penetrates the RF shield then the RF Shield Vendor must install the anchors prior to system equipment delivery. The need for Blower Box anchors must be planned in advance to allow the screen room vendor to include the cost in their bid.

7–7–4 Physical Characteristics (Continued)

Material Properties For The Stud And Anchor

The stud must be made of a conductive material to allow it to be electrically connected to the Room's RF Shield at the point of penetration. The material properties should be compatible with the material properties of the RF Shield and not produce galvanic corrosion due to dis–similar metals. The stud does not require electrical isolation from the equipment base.

The embedment anchor must comply with local building codes especially for seismic designated areas. An epoxy type anchor may be used if it meets the following requirements:

1. Able to achieve clamping force (tension) requirement for the equipment to be anchored.

2. Is approved by the local building inspector.

7–7–5 Installation Location

The exact location for installing the Magnet anchors is determined by dimensional footprint drawings for the MR equipment to be installed. The Installation Services Group at GE Medical Systems will provide to the RF Shield Room Vendor the dimensional drawing showing all anchor locations. The drawing can be issued in either hard copy or electronic format. The RF Shield Room Vendor is responsible for supplying their own template to precisely mark the Magnet anchor locations within the room.

GE Service to attach/align Dock Assembly to Magnet and mark anchor location for RF Shield Room Vendor.

Coordination between the RF Shield Room Vendor and Building Contractor/Architect may be necessary to mark the location of the anchors to prevent interference with rebar or structural steel. A re–arrangement of the room may be necessary to ensure ground isolation.

Refer to Table 7–6, equipment dimensional illustration references.

7–7–6 Example – Select Magnet Anchor Size

The following is an example of to illustration the selecting of proper anchor to install an LCC Magnet into a building structure with 2000 psi (13.8 MPa) concrete. For this example the area is not under seismic requirements.

1. Determine magnet clamping force (tension) by referring to requirements in Table 7–5.

2500 lbs + 200 lbs = 2700 lbs	11,100 N + 900 N = 12,000 N

2. Refer to Tables 7–7 or 7–8 (examples of anchor vendor catalogs) to select anchor diameter and embedment which meets the clamping force (tension) determined in Step 1.

Diameter : Min. 0.625 inch Max. 1.25 inch	Diameter : Min. M16 Max. M32
For 8 inch embedment select 3/4 inch diameter	For 130 mm embedment select M20 diameter
For 4.5 inch embedment select 1 inch diameter	For 114 mm embedment select M24 diameter

3. The vendor instructions and torque to the maximum recommended level for the anchor selected in Step 2 must be provided to the RF Shield Room vendor for proper installation of the anchor and equipment.

7–7–6 Example – Select Magnet Anchor Size (Continued)

TABLE 7–7
EXAMPLE OF ENGLISH UNITS STAINLESS STEEL ANCHOR ALLOWABLE LOADS IN CONCRETE

ANCHOR DIAMETER in. (mm) See Note 1	EMBEDMENT DEPTH in. (mm)	2000 psi (13.8 MPa)		3000 psi (20.7 MPa)		4000 psi (27.6 MPa)		6000 psi (41.4 MPa)	
		TENSION lb (kN)	SHEAR lb (kN)	TENSION lb (kN)	SHEAR lb (kN)	TENSION lb (kN)	SHEAR lb (kN)	TENSION lb (kN)	SHEAR lb (kN)
5/8 (15.9)	2 3/4 (70)	1250 (5.6)	2800 (12.5)	1600 (7.1)	3070 (13.7)	1810 (8.1)	3330 (14.8)	1920 (8.5)	3330 (12.5)
	4 (102)	1870 (8.3)	3330 (14.8)	2400 (10.7)	3330 (14.8)	2930 (13.0)	3330 (14.8)	3200 (14.2)	3330 (12.5)
	7 (178)	2500 (11.2)	3330 (14.8)	3010 (13.4)	3330 (14.8)	3650 (16.2)	3330 (14.8)	3650 (16.2)	3330 (12.5)
3/4 (19.1)	3 1/4 (83)	1550 (6.9)	2880 (12.8)	1950 (8.7)	3310 (14.7)	2350 (10.5)	3730 (16.6)	2610 (11.6)	4800 (21.4)
	4 3/4 (121)	2510 (11.2)	4510 (20.1)	3250 (14.5)	4650 (20.7)	3870 (17.2)	4800 (21.4)	4670 (20.8)	4800 (21.4)
	8 (203)	2930 (13.0)	4800 (21.4)	3870 (17.2)	4800 (21.4)	4530 (20.2)	4800 (21.4)	5120 (22.8)	4800 (21.4)
1 (25.4)	4 1/2 (114)	3120 (13.9)	6080 (27.0)	3870 (17.2)	6770 (30.1)	4610 (20.5)	7470 (33.2)	4800 (21.4)	7470 (33.2)
	6 (152)	4400 (19.6)	7470 (33.2)	6400 (28.5)	7470 (33.2)	7200 (32.0)	7470 (33.2)	7330 (32.6)	7470 (33.2)
	9 (229)	5600 (24.9)	7470 (33.2)	8000 (35.59)	7470 (33.2)	9390 (41.77)	7470 (33.2)	9390 (41.8)	7470 (33.2)
Note 1 All shaded values fail to meet the clamping force (tension), therefore are not acceptable anchors.									

TABLE 7–8
EXAMPLE OF METRIC STAINLESS STEEL ANCHOR ALLOWABLE LOADS IN CONCRETE

ANCHOR DIAMETER See Note 1	EMBEDMENT DEPTH mm (in.)	13.8 MPa (2000 psi)		20.7 MPa (3000 psi)		27.6 MPa (4000 psi)		41.4 MPa (6000 psi)	
		TENSION kN (lb)	SHEAR kN (lb)	TENSION kN (lb)	SHEAR kN (lb)	TENSION kN (lb)	SHEAR kN (lb)	TENSION kN (lb)	SHEAR kN (lb)
M16	105 (4 1/8)	11.2 (2500)	25.1 (5650)	20.9 (4705)	39.9 (8965)	24.2 (5450)	10125 (45.0)	6900 (30.7)	10550 (46.9)
M20	130 (5 1/8)	25.1 (5650)	52.9 (11900)	30.7 (6910)	58.7 (13195)	36.4 (8175)	14490 (64.5)	10005 (44.5)	14490 (64.5)
M24	155 (6 1/8)	30.0 (6735)	61.2 (13760)	36.9 (8300)	70.5 (15855)	43.9 (9860)	29.8 (17950)	57.7 (12980)	95.6 (21490)
Note 1 All shaded values fail to meet the clamping force (tension), therefore are not acceptable anchors.									

7–8 MAGNET ROOM EQUIPMENT MOUNTING

7–8–1 Remote Oxygen Sensor Module (OM3) – Optional

The Remote Oxygen Sensor Module (if option ordered) must be mounted approximately 60 in. (1524 mm) above the Magnet Room floor near the front of the magnet enclosure.

7–8–2 Magnet Rundown Unit (MS4)

The magnet rundown unit should be mounted 60 in. (1524 mm) above the Magnet Room floor near the front of the magnet enclosure but outside the 200 gauss zone.

7–8–3 Emergency Off buttons

Customer supplied emergency off buttons to be located near each room exit including magnet and equipment rooms. These buttons must be clearly labeled, “Emergency Off”. Refer to Section 5, POWER REQUIREMENTS, for emergency off power requirements.

7–9 RF DOOR SWITCH

RF shielded room vendor must supply and install RF door switches on all RF shielded doors. These switches must be wired in series and a GE supplied cable (two loose lead conductors) will attach to one door switch. RF switches must be rated for 24 volts at 750 milliamperes maximum and the switches must be in the open position when the doors are open (switch contacts close when the doors are completely closed).

7–10 EMERGENCY EXIT

Emergency exiting from the Magnet Room is to be specified by the customer’s architect and contractor. Such measures as an out swinging door, emergency door latch release, easily removed window, or other measures must be designed into the room. Emergency exit instructions must be permanently and prominently mounted near the door and/or window.

7–11 ROOM VENTILATION SWITCH

Placement of the room ventilation switch should be near the Magnet Room door and is the responsibility of the architect and mechanical contractor.